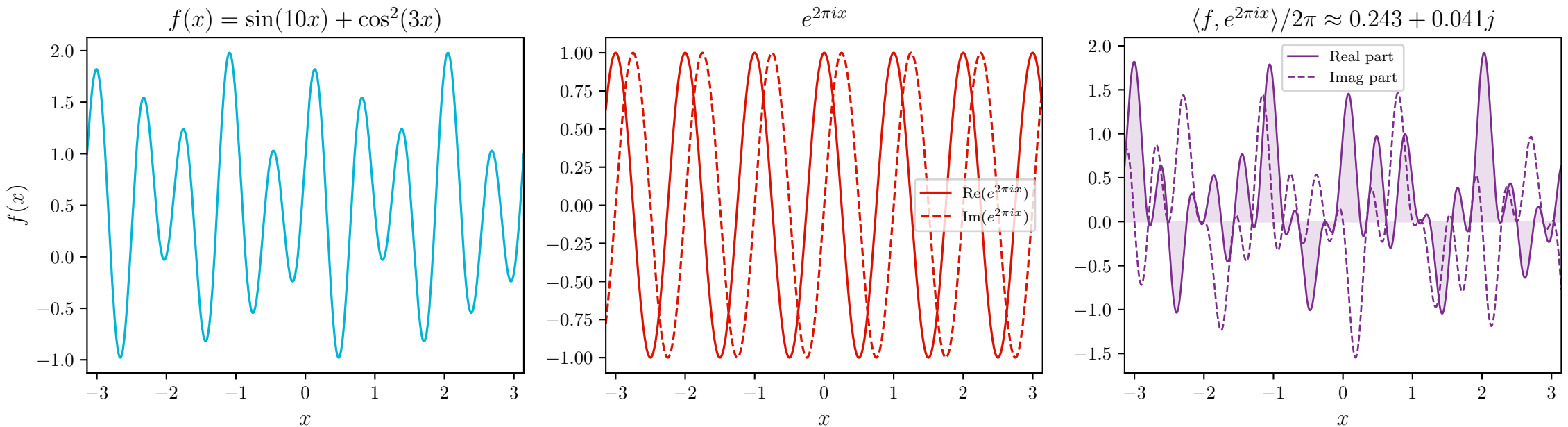
$f(x) = \sin(10x) + \cos^2(3x)$
$f(x)$
x
$e^{2\pi i x}$
$\mathrm{Re}(e^{2\pi i x})$
$\mathrm{Im}(e^{2\pi i x})$
x
$\langle f, e^{2\pi i x}\rangle / 2\pi \approx 0.243 + 0.041j$
Real part
Imag part
x